unsigned int	An integer without any sign (minus symbol)	2 bytes	16 bits	0 to 65535
signed int	An integer with sign	2 bytes	16 bits	-32,768 to 32,767
Both short and int are similar				
long	long is short name for long int	4 bytes	32 bits	-2147483648 to 2147483647
unsigned long	A double spaced integer without any sign	4 bytes	32 bits	0 to 4,294,967,295
signed long	A double spaced integer with sign	4 bytes	32 bits	-2147483648 to 2147483647

The above table clearly shows that an integer type accepts only 2 bytes of data whereas a long int accepts data that is double this size i.e., 4 bytes of data. So, we can store more digits in a long int. (long is a modifier and int is a fundamental data type)

char type

Table 9.7 Memory allocation for char Data types

Data type		Space in memory		Range of value
		in terms of bytes	in terms of bits	
char	Signed ASCII character	1 byte	8 bits	-128 to 127
unsigned char	ASCII character without sign	1 byte	8 bits	0 to 255
signed char	ASCII character with sign	1 byte	8 bits	-128 to 127

Floating point type

Table 9.8 Memory allocation for floating point Data types

Data type		Space in memory		Range of value
		in terms of bytes	in terms of bits	
float	signed fractional value	4 bytes	32 bits	3.4×10^{-38} to $3.4\times10^{38}-1$
double	signed more precision fractional value	8 bytes	64 bits	1.7×10^{-308} to $1.7\times10^{308}-1$
long double	signed more precision fractional value	10 bytes	80 bits	3.4×10^{-4932} to $1.1\times10^{4932}-1$

Memory allocation is subjected to vary based on the type of compiler that is being used. Here, the given values are as per the **Turbo C++** compiler. **Dev C++** provides some more space to **int** and **long** double types. Following Tables 9.9 shows the difference between Turbo C++ and Dev C++ allocation of memory.

Table 9.9 Memory allocation of Turbo C++ and Dev C++

11th Std Computer Science_EM Chapter 9.indd 137 06-01-2023 15:08:46